

Commit e8f8d3d

Browse files

ActuallyNull committed on Aug 9

finished physionet X, y dataset loader, still need mit-bih loader

main 1 parent 0f43883 commit e8f8d3d

Filter files...

- data_processing.py

testingScripts

ecg_annotations_test.py

ecg_processor_test.py

3 files changed +40 -12 lines changed

Search within code

data_processing.py +31 -9

@@ -2,20 +2,42 @@

2 2 import wfdb

3 3 import numpy as np

4 4 import os

5 + import pandas as pd

5 6

6 7 dataset_folders = ["mit-bih", "training2017/training2017"]

7 8 records = []

9 + labels = []

10 + physionet_label_map = {'N': "Normal", 'A': "AFib", 'O': "Other Arrythmia"}

8 11

9 - for folder in dataset_folders:

10 - for file in os.listdir(folder):

11 - if file.endswith(".hea"): # .hea is a common extension in all dataset_folders

12 - record_name = file[:-4] #splicing (removing) extension name

12 + X = []

13 + y = []

14 +

15 + def load_physionet_dataset(physionet_path, reference_file_path):

16 +

17 + reference_df = pd.read_csv(reference_file_path, header=None, names=['record', 'label'])

18 +

19 + # loading the data

20 + for file in os.listdir(physionet_path):

21 + if file.endswith(".hea"):

22 + record_name = file[:-4]

13 23 records.append(record_name)

14 24

15 - X = []

25 + for name in records:

26 + nameLabel = reference_df.loc[reference_df['record'] == name, 'label'].values[0]

27 + if nameLabel == "N":

28 + labels.append("Normal")

29 + elif nameLabel == "A":

30 + labels.append("AFib")

31 + else:

32 + labels.append("Other Arrythmia")

33 +

34 + # entering it in dataset

35 + for paths in records:

36 + record = wfdb.rdrecord(physionet_path+"\\\\"+paths)

37 + ecg_signal = record.p_signal[:, 0]

38 + X.append(ecg_signal)

16 39

17 - for paths in records:

18 - record = wfdb.rdrecord(paths)

19 - ecg_signal = record.p_signal[:, 0]

20 - X.append(ecg_signal)

40 + for label in labels:

41 + y.append(label)

21 42

43 + load_physionet_dataset("training2017\\training2017", "training2017\\training2017\\REFERENCE.csv")

testingScripts/ecg_annotations_test.py +5

@@ -0,0 +1,5 @@

1 + import wfdb

2 +

3 + annotations = wfdb.rdann("mit-bih\\100", "atr")

4 +

5 + print(annotations.symbol)

ecg_processor_test.py testingScripts/ecg_processor_test.py +4 -3

@@ -8,7 +8,8 @@

8 8 ecg_signal = record.p_signal[:, 0]

9 9 duration = 5

10 10 samples = fs * duration

11 + print(record.comments)

11 12

12 - _, rpeaks = nk.ecg_peaks(ecg_signal, sampling_rate=fs)

13 - plot = nk.events_plot(rpeaks['ECG_R_Peaks'][:5], ecg_signal[:6*fs])

14 - plot.savefig("ecg.png")

13 + #_, rpeaks = nk.ecg_peaks(ecg_signal, sampling_rate=fs)

14 + #plot = nk.events_plot(rpeaks['ECG_R_Peaks'][:5], ecg_signal[:6*fs])

15 + #plot.savefig("ecg.png")

Comments 0

Please sign in to comment.